

SYDE 556/750

Simulating Neurobiological Systems
Lecture 5: Feed-Forward Transformation

Chris Eliasmith

Oct 1 & 6, 2026

- ▶ Slide design: Andreas Stöckel
- ▶ Content: Terry Stewart, Andreas Stöckel, Chris Eliasmith



UNIVERSITY OF
WATERLOO

FACULTY OF
ENGINEERING



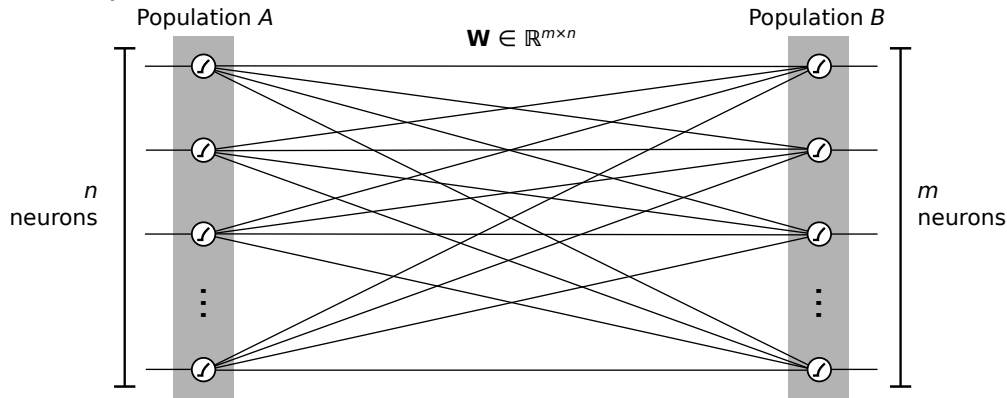
Introduction

- ▶ We've only talked about representation til now
 - ▶ What about computation?
- ▶ We start by focusing on the state of a network after learning and development
- ▶ A kind of hypothesis testing and generation



DALL-E AI Generated Art, 2022

NEF Principle 2: Transformation

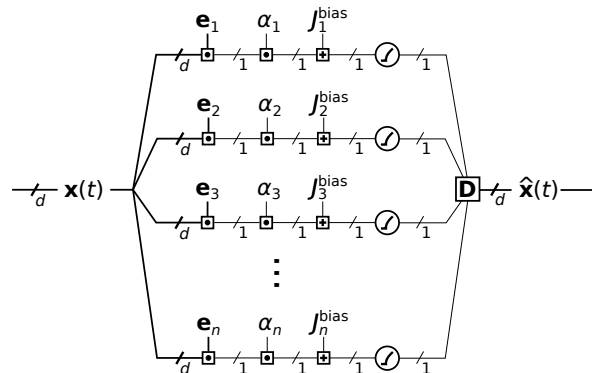


NEF Principle 2 – Transformation

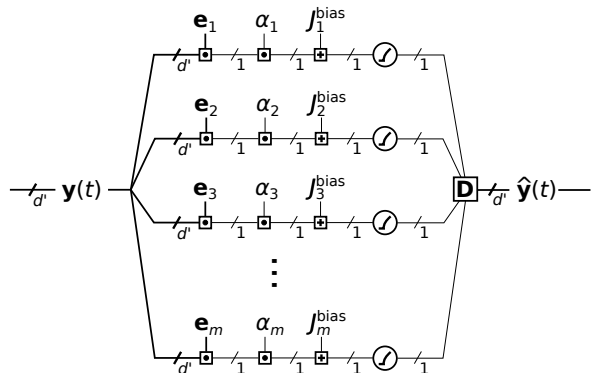
Connections between populations describe *transformations* of neural representations. Transformations are functions of the variables represented by neural populations.

A Tale of Two Populations (I)

Population A

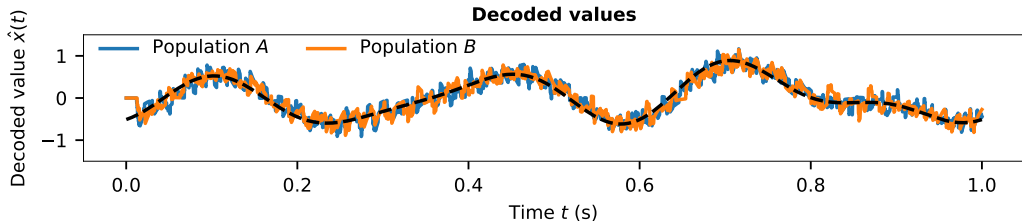
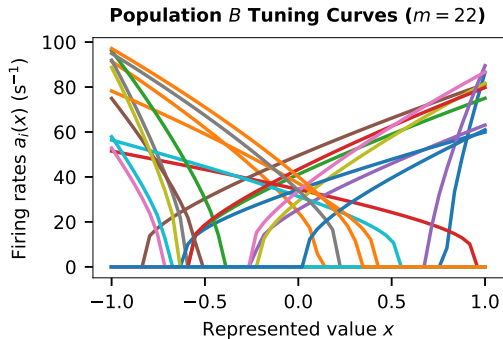
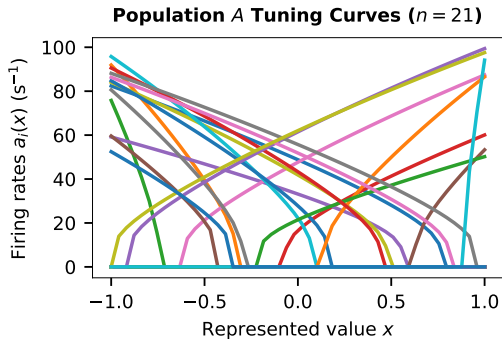


Population B

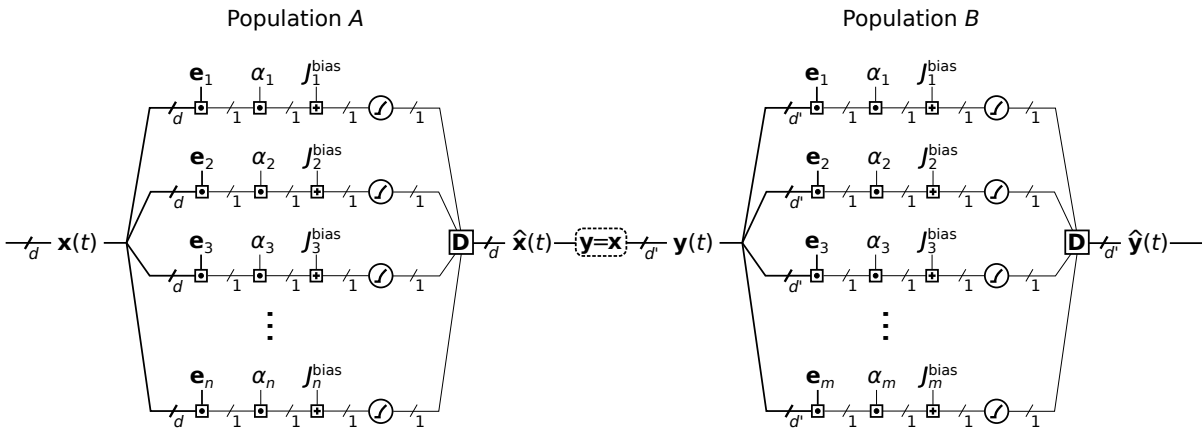


Filters?

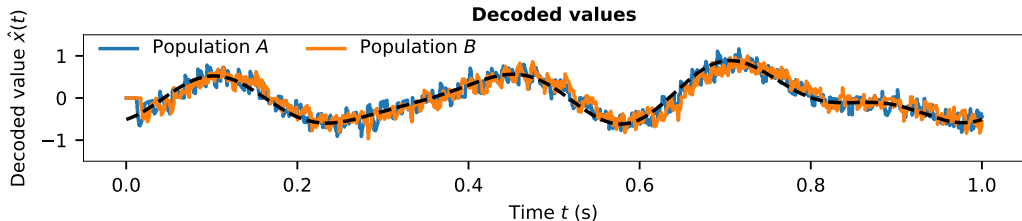
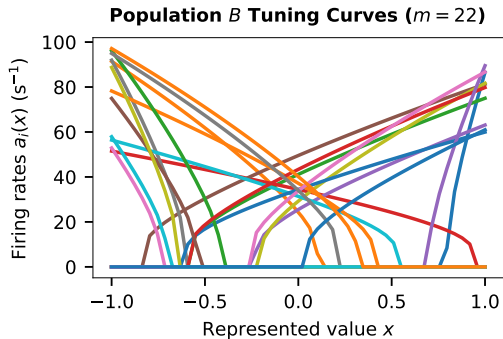
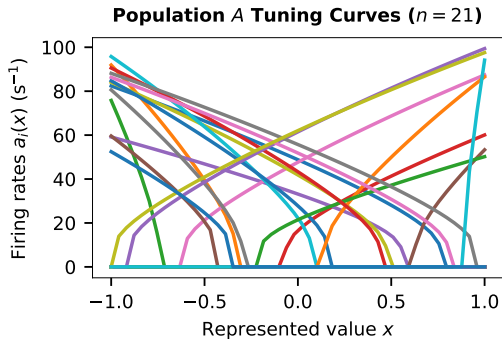
Communication Channel Experiment: Same input signal



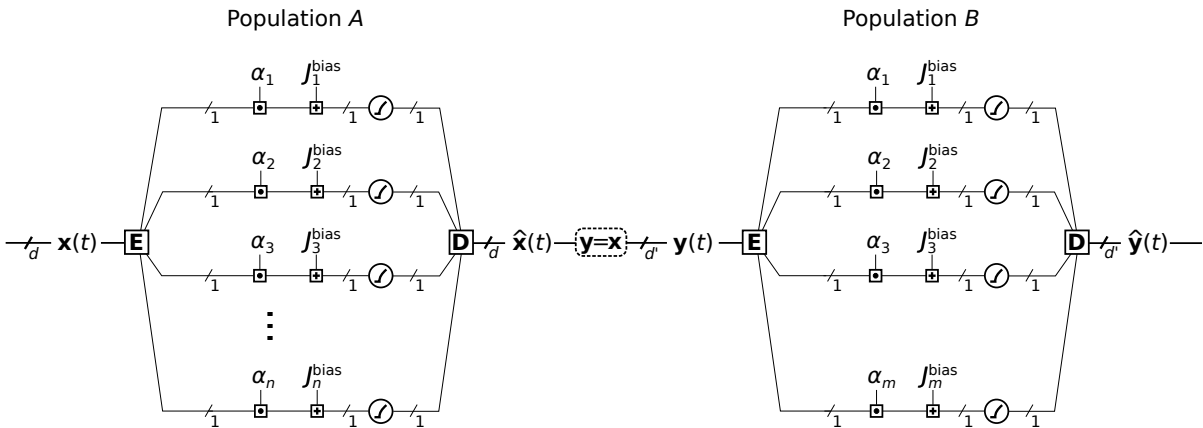
A Tale of Two Populations (II)



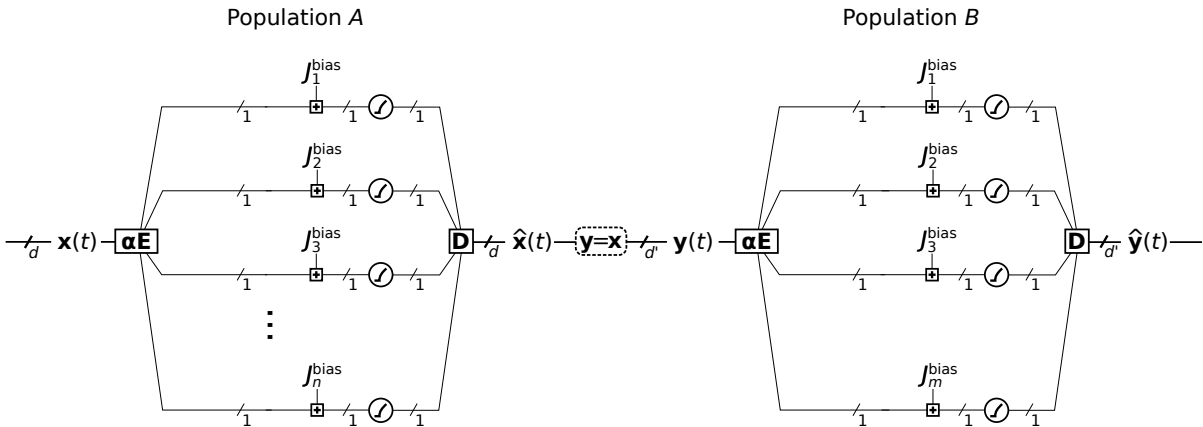
Communication Channel Experiment: Populations in series



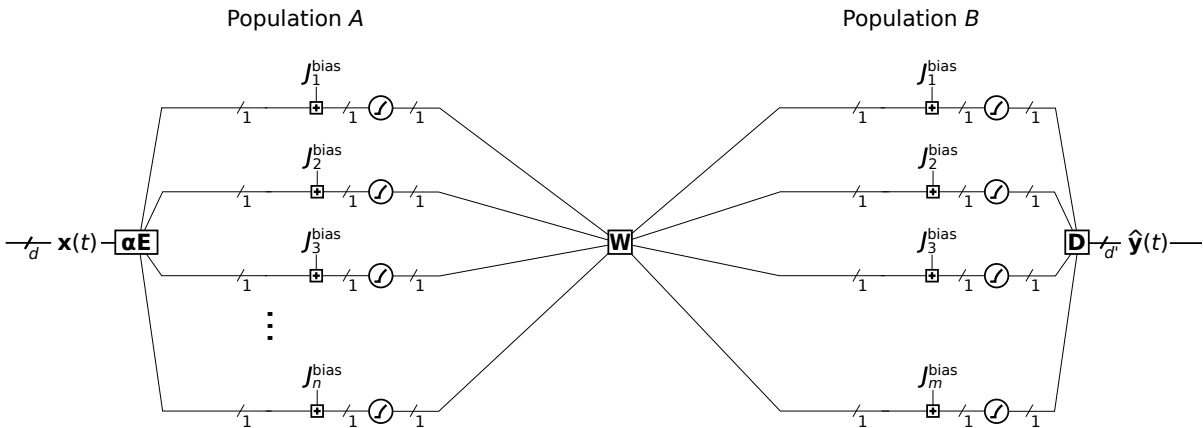
Computing Synaptic Weights: Step 1 – Encoding Matrix



Computing Synaptic Weights: Step 2 – Scaled Encoding Matrix



Computing Synaptic Weights: Step 3 – $\mathbf{W} = \mathbf{ED}$

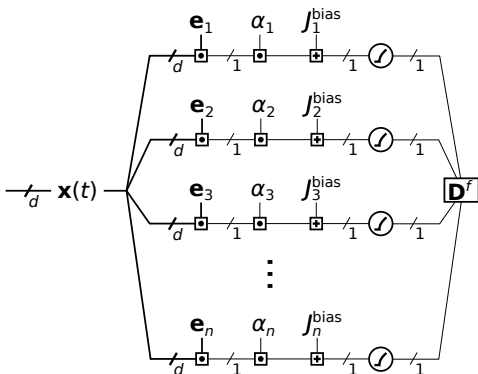


Computational Complexity

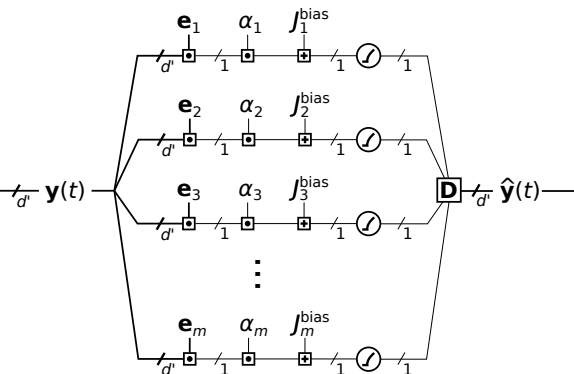
- ▶ Weights - multiplying $\mathbf{a} \in \mathbb{R}^n$ with $\mathbf{W} \in \mathbb{R}^{m \times n}$ is $\mathcal{O}(nm)$ i.e., $\approx \mathcal{O}(n^2)$
- ▶ Decoding - $\hat{\mathbf{x}} = \mathbf{D}\mathbf{a}$ is $\mathcal{O}(dn)$
- ▶ Encoding - $\mathbf{J} = \mathbf{E}\hat{\mathbf{x}} + \mathbf{J}_{\text{bias}}$ is $\mathcal{O}(dm)$
- ▶ Encoding/Decoding - $\mathcal{O}(d(n + m))$ or $\approx \mathcal{O}(dn)$ for $n = m$
- ▶ So if d is small we get a linear complexity $\mathcal{O}(n)$
- ▶ Therefore, sequential decoding and re-encoding saves a lot of time compared to using actual synaptic weights
- ▶ One reason why Nengo is so fast compared to other SNN simulators

Computing Functions

Population A

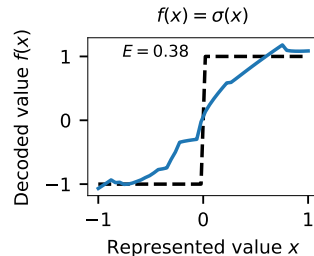
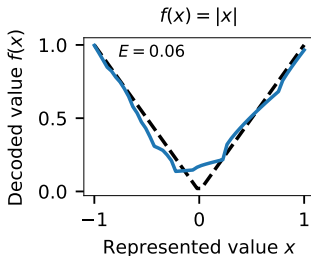
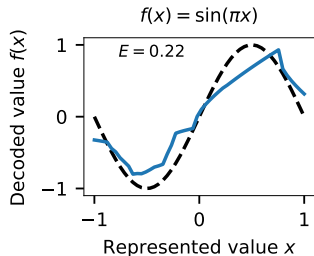
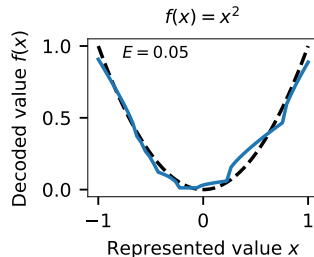
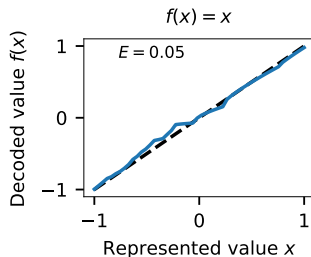
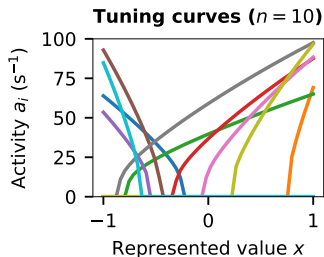


Population B



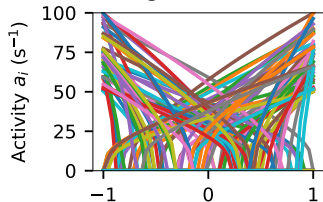
Function Decoder $\mathbf{D}^f = ((\mathbf{A}\mathbf{A}^\top + N\sigma^2\mathbf{I})^{-1}\mathbf{A}\mathbf{Y}^\top)^\top$, where $(\mathbf{Y})_{ik} = (f(\mathbf{x}_k))_i$

Decoding Functions – Using a Few Neurons



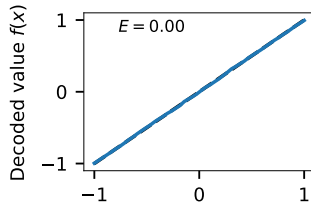
Decoding Functions – Using More Neurons

Tuning curves ($n = 100$)



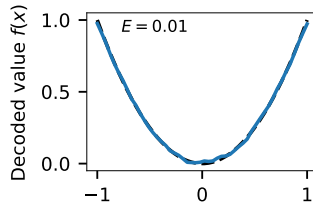
Represented value x

$$f(x) = x$$



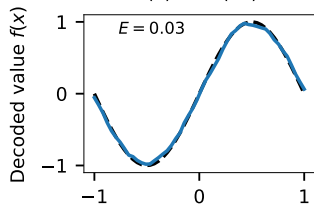
Represented value x

$$f(x) = x^2$$



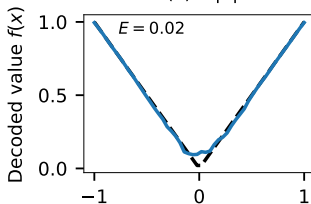
Represented value x

$$f(x) = \sin(\pi x)$$



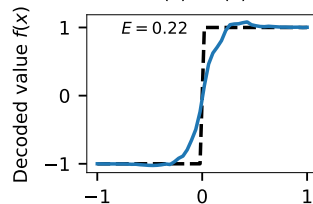
Represented value x

$$f(x) = |x|$$



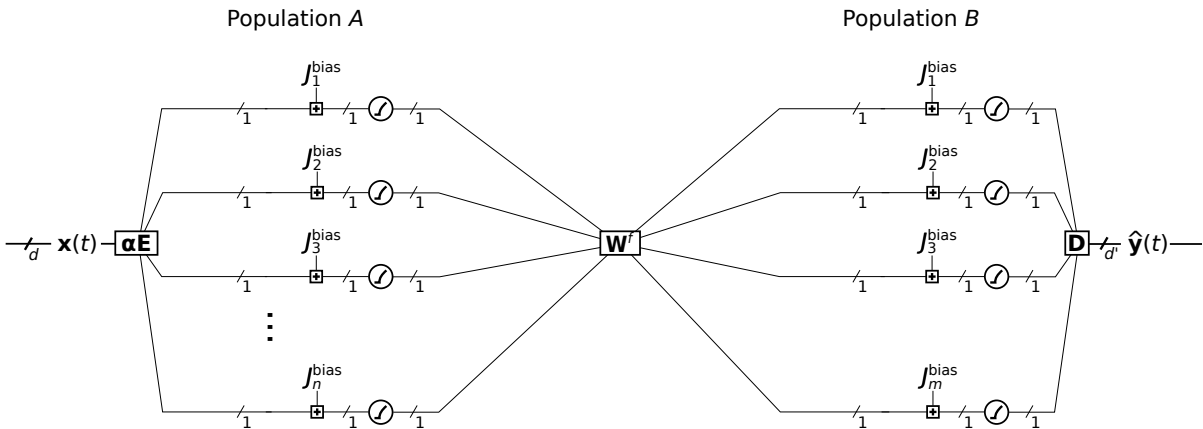
Represented value x

$$f(x) = \sigma(x)$$



Represented value x

Computing Functions – Weight Matrix



$$W^f = E D^f$$

Recipe for any feedforward transformation

1. Define encoding for two populations (input/output)
2. Write the transformation with the represented input variables
3. Solve for the $\mathbf{D}^f$ for that transformation for the input population
4. Sub 3. into the encoding for the output variable

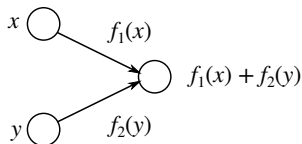
Computing Multivariate Functions

○ Homogenous population ⊗ Heterogenous population

→ Linear connection —| Inh. connection —● Exc. connection

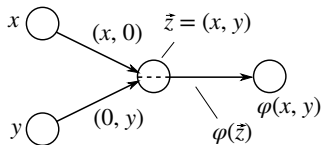
Linear Superposition

$$W^{f_1} \mathbf{a}_1(\mathbf{x}) + W^{f_2} \mathbf{a}_2(\mathbf{y})$$



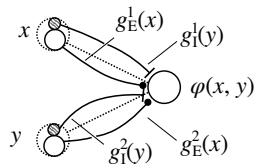
Nonlinear Functions

Multi-dimensional $\mathbf{z}$



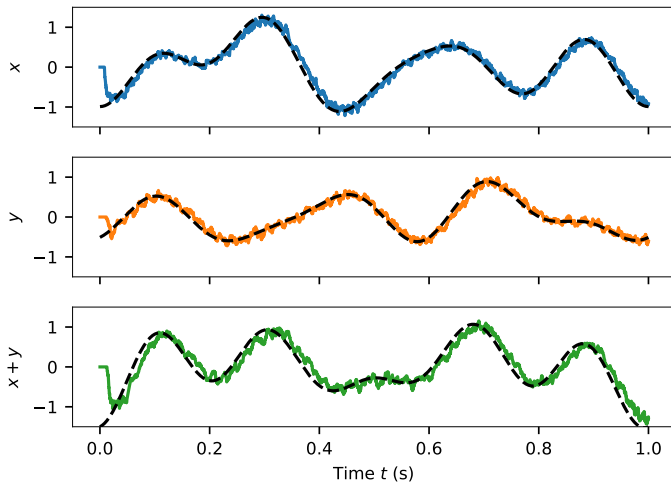
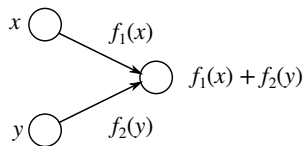
(Dendritic Computation)

Exploit dendritic nonlinearity



Computing Multivariate Functions – Linear Superposition

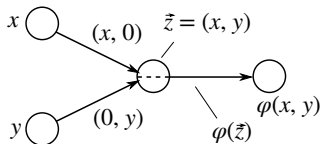
Linear Superposition



Computing Multivariate Functions – Multiplication

Nonlinear Functions

Multi-dimensional $\mathbf{z}$



Multiplication is useful...

- Gating of signals
- Attention effects
- Binding
- Statistical inference

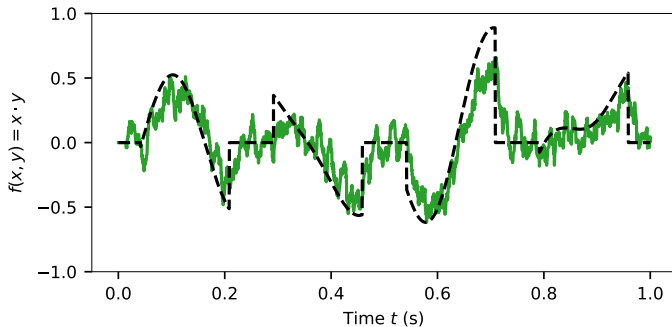
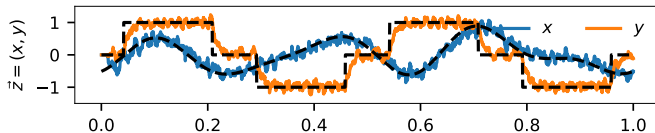


Image sources

Title slide

“Yellow Butterfly”

Author: Albert Bierstadt, circa 1890.

From Wikimedia.